

2. Deployment Options

Reference: [install/before-you-start/deployment-options/](#)

2.1 Choosing a Platform

Supported host platforms for CAST Imaging

Platform	Description	Best fit for
Windows Server (native)	Traditional install using the CAST Imaging Windows setup package.	Organizations standardized on Windows infrastructure, no container platform available.
Docker	CAST Imaging distributed as a set of Docker containers orchestrated via Docker Compose.	Linux-first shops, fastest path to a working install, POCs, small/medium teams.
Podman	Daemonless, rootless-capable container engine; drop-in alternative to Docker for security-hardened Linux environments.	Environments where Docker daemon (root-owned) is not permitted (e.g. RHEL-based, government/regulated).
Kubernetes	Helm-chart / manifest-based deployment onto an existing K8s cluster.	Enterprises with an established container orchestration platform, need for high availability, autoscaling, and centralized ops.

2.2 Choosing a Scenario

Evaluation / POC

Fastest setup, minimal hardening, single machine, self-signed certs acceptable, local authentication.

Standard Production

Hardened OS, HTTPS with CA-signed certificate, SSO (SAML/LDAP), backups, monitoring, dedicated DB.

Enterprise / High Availability

Multi-machine or Kubernetes, redundancy at web/app tier, managed/clustered database,

centralized log/metric collection, change-managed upgrades.

2.3 Choosing a Topology (Single vs. Multi-Machine)

Topology decision matrix by situation

Situation	Recommended Topology	Recommended Platform	Rationale
Proof of Concept / trial, <~50 applications	Single machine (all-in-one)	Docker / Podman	Minimizes setup effort; all services (web, application, database) co-located on one host or one docker-compose stack.
Small team, 1 project, <~150 applications	Single machine, or multi-machine with a dedicated DB host	Docker / Podman / Windows	Single machine is usually sufficient; separate the database early if growth is expected.
Multiple teams / departments, shared platform	Multi-machine (dedicated web/app node + dedicated DB node)	Docker / Podman / Kubernetes	Isolates DB I/O from web/app workload; simplifies backup and scaling of the database independently.
Enterprise-wide rollout, HA / SLA requirements	Multi-machine, redundant web/app tier	Kubernetes (preferred) or clustered Docker/Podman hosts behind a load balancer	Enables rolling upgrades, horizontal scaling, and failover without downtime.
Regulated / air-gapped environment	Single or multi-machine, offline install bundle	Podman (rootless) or Windows	Avoids Docker daemon root requirement; supports offline/registry-mirrored image delivery.

Rule of thumb. Start single-machine for POCs and small teams; move the database to its own host (or managed service) as soon as the deployment becomes production-facing or exceeds ~150 delivered applications, or when multiple concurrent users/CI pipelines push load onto the database.

3.1 Hardware Requirements

Reference: [install/requirements/hardware/](#) — refined by the topology chosen in Section 2.

Hardware sizing by scenario and topology (see also [sizing-requirements-matrix.csv](#))

Scenario	Topology	Component	vCPU	RAM	Disk
POC / Evaluation	Single machine	All-in-one node	4	16 GB	150 GB SSD
Small Team	Single machine	All-in-one node	8	32 GB	300 GB SSD
Small Team	Multi-machine	Web/Application node	4	16 GB	100 GB SSD
Small Team	Multi-machine	Database node	4	16 GB	200 GB SSD
Standard Production	Multi-machine	Web/Application node	8	32 GB	200 GB SSD
Standard Production	Multi-machine	Database node	8	32 GB	500 GB+ SSD
Enterprise / HA	Multi-machine / Kubernetes	Web/App tier (replicated)	16 (aggregate)	64 GB (aggregate)	400 GB SSD
Enterprise / HA	Multi-machine / Kubernetes	Database tier	16	64 GB	1 TB+ SSD, clustered/managed

Sizing is workload-dependent. The effective RAM/CPU/disk needed scales with the number and size of delivered applications, retained snapshots, concurrent analyses, and number of active users. Treat the table above as a starting point and monitor utilization after go-live to right-size.

General hardware guidance

- SSD (or equivalent low-latency block storage) is strongly recommended for the database tier in all scenarios.

- CPU should favor higher single-core performance for the analysis/indexing engine; RAM sizing should account for concurrent analyses.
- Reserve headroom (20–30%) above the minimum for OS, container runtime, and monitoring agents.
- Virtualized hardware (VMware/Hyper-V/cloud VM) is fully supported; avoid CPU/RAM oversubscription on the database host.

3.2 Software Requirements

Reference: [install/requirements/software/](#)

Operating Systems

Family	Supported versions (typical)	Notes
Windows	Windows Server 2019 / 2022 (64-bit)	For native Windows installation of CAST Imaging.
Linux (Docker/Podman host)	RHEL 8/9, Ubuntu Server LTS (20.04/22.04), other Docker/Podman-certified distributions	Kernel and container-runtime compatibility should match the Docker/Podman version requirement below.
Kubernetes nodes	Any CNCF-conformant, currently supported Kubernetes minor version	Cluster must already exist; CAST Imaging is deployed onto it, not used to provision it.

Container Runtime & Orchestration

Component	Minimum version (indicative)
Docker Engine + Docker Compose	Recent supported Docker Engine (20.10+) and Compose v2
Podman + podman-compose / Podman Kube	Podman 4.x+
Kubernetes	Current CNCF-supported release; Helm 3.x if chart-based install is used

Client / End-User Software

- **Web browser** (Imaging Console / Viewer / dashboards): latest two major versions of Google Chrome, Microsoft Edge (Chromium), or Mozilla Firefox.

- **AIP Console / delivery tooling** (where applications are prepared prior to delivery): requires a compatible JRE and, depending on technology analyzed, source control clients (Git, SVN, TFS/Azure DevOps clients).
- No CAST client software installation is required for end users who only consult results through the browser-based Console/Viewer.

Always confirm exact version numbers. CAST publishes a compatibility matrix per release on doc.castsoftware.com/imaging/install/requirements/software/ ; minimum OS/Docker/Kubernetes versions are revised with each CAST Imaging release.

3.3 Database Requirements

Reference: [install/requirements/db/](#)

Item	Requirement
Primary supported engine	PostgreSQL (bundled as a container in Docker/Podman/Kubernetes deployments, or externally hosted/managed for production)
Alternate supported engine	Microsoft SQL Server (where supported by the specific CAST Imaging release — verify current compatibility matrix)
Character set / collation	UTF-8 encoding required; case-sensitive collation as specified by CAST documentation
Dedicated instance	Recommended in Standard/Enterprise scenarios — do not share the instance with unrelated production workloads
Backup & retention	Regular automated backups required; database growth tracks number of delivered applications, snapshot history and dashboard data
High availability	For Enterprise/HA topologies, use a managed or clustered PostgreSQL service (e.g. streaming replication, managed cloud DBaaS)
Network access	TCP 5432 (PostgreSQL) or TCP 1433 (SQL Server) reachable from every Imaging application node

Sizing depends on scale. Database disk sizing must be refined against the number of applications, the size of each Knowledge Base, and how long historical snapshots are retained — see the Disk Requirements section for storage-location detail.

3.4 Disk Requirements & Storage Locations

Reference: [install/requirements/disk/](#) and [install/requirements/disk/storage-locations/](#)

Primary storage locations and sizing drivers

Storage Location	Purpose	Sizing Driver	Growth Pattern
Installation directory (binaries/containers/images)	Application and container image files	Fixed per version	Grows only on upgrade (keep N-1 version during rollback window)
Database data directory	PostgreSQL/SQL Server data files	Number of applications × Knowledge Base size × retained snapshots	Continuous growth — plan quarterly review
Application data / delivery workspace	Staging area for delivered application packages before/while processed	Largest single application package size × concurrent deliveries	Transient, but must be sized for peak concurrency
Logs directory	Application, web server, and container logs	Log verbosity × retention window	Steady growth; rotate/ship to central logging
Backup / export location	Database backups, exported reports	DB size × retention count	Grows with backup retention policy
Docker/Podman volumes	Persistent volumes backing the above when containerized	Same as underlying data type	Must be on a persistent, non- ephemeral disk/volume

- Use separate volumes/partitions for database data vs. logs vs. delivery workspace so that a runaway log or workspace cannot fill the DB disk.
- All persistent storage locations (DB, volumes) must survive container/host restarts — never place them on ephemeral/local-only container filesystem layers.

- In Kubernetes, back persistent locations with a durable `PersistentVolume` / storage class (not `emptyDir`).

3.5 Server & End-User Computer Requirements

Server-Side (hosting CAST Imaging)

- Meets the hardware/software/DB requirements in Sections 3.1–3.3 for the chosen scenario.
- Static IP / stable DNS name resolvable both internally and (if externally published) externally.
- NTP-synchronized clock (required for valid TLS certificate checks and consistent log timestamps).
- Outbound HTTPS access per Section 9 (extend.castsoftware.com, optional Highlight/LLM endpoints).

End-User Workstation (consulting results)

Item	Minimum
Browser	Latest 2 major versions of Chrome, Edge, or Firefox
RAM	8 GB (for smooth rendering of large architecture graphs / dashboards)
Network	Stable connectivity to the CAST Imaging web endpoint (HTTPS 443); low latency improves graph interaction responsiveness
Screen resolution	1920×1080 or higher recommended for architecture/dependency graph exploration

Delivery / Analyst Workstation (preparing & delivering applications)

- Same as end-user workstation, plus: local disk space for source-code checkouts, a compatible JRE, and any source-control client tools needed (Git/SVN/TFS).
- Network access to internal source-code repositories (Section 9) and to the CAST Imaging delivery endpoint.

4. MCP Servers Overview

Reference: [imaging/mcp-server/](#) , [mcp-server/imaging/](#) , [mcp-server/gatekeeper/](#) , [mcp-server/oauth/](#)

CAST Imaging exposes its knowledge base to AI assistants and coding agents via the **Model Context Protocol (MCP)**. This introduces additional components and network flows beyond the core web/application/database stack.

CAST Imaging MCP Server

Exposes CAST Imaging's application knowledge (architecture, dependencies, transactions) as MCP tools/resources so AI assistants (e.g. IDE copilots, chat agents) can query it in natural language.

Gatekeeper

Acts as an authorization/policy front-door in front of the MCP server(s), enforcing which clients/users can call which MCP tools and against which applications.

OAuth Module

Provides OAuth2 / OIDC-based authentication for MCP clients, so AI assistants authenticate as the requesting user rather than with a shared static secret.

Prerequisites specific to MCP Servers

- A running, reachable CAST Imaging instance (the MCP server is a companion service, not a replacement for the core platform).
- An OAuth2/OIDC provider (built-in, or the same IdP used for SAML/LDAP SSO) reachable from both the MCP Gatekeeper and the end-user's AI client.
- Outbound HTTPS (443) from the MCP server host to the configured LLM API endpoint, if AI-generated natural-language responses are used (see Section 9).
- Inbound connectivity from AI client tooling (IDE plugin, chat client, orchestration agent) to the MCP server/Gatekeeper endpoint — typically HTTPS 443, proxied the same way as the main Imaging web UI.
- Sizing: lightweight relative to the core platform — see [sizing-requirements-matrix.csv](#) (2 vCPU / 4 GB RAM indicative per MCP service).

Security note. Because MCP tools can expose detailed architecture, data-flow, and dependency information (and, via Gatekeeper, potentially broader actions), always front the MCP endpoint with the OAuth module and Gatekeeper policy enforcement in any environment beyond an isolated POC, and restrict outbound LLM traffic to the exact FQDNs required by the chosen AI provider (see Section 9).

5. Installation Reference

Reference: [install/global/](#), [install/global/windows/](#), [install/global/docker/](#), [install/global/podman/](#), [install/global/kubernetes/](#)

Platform	Prerequisite checklist (high level)	Typical install artifact
Windows	Windows Server meeting hardware/software requirements; local administrator rights; .NET/JRE prerequisite installed; database reachable; firewall rules opened (Section 10)	CAST Imaging setup executable / MSI
Docker	Docker Engine + Compose installed; user in <code>docker</code> group or sudo access; outbound access to pull images; persistent volumes provisioned; <code>docker-compose.yml</code> / env file configured with DB, license, and HTTPS settings	Docker Compose stack (multi-container)
Podman	Podman + podman-compose (or Podman Kube YAML) installed; rootless configuration validated if used; SELinux contexts configured for volumes on RHEL-family hosts	Podman Compose / Kube YAML stack
Kubernetes	Existing supported cluster; Ingress controller for HTTPS termination or pass-through; StorageClass for persistent volumes; Helm 3.x (if chart-based); namespace and RBAC prepared; secrets for DB credentials/license pre-created	Helm chart / Kubernetes manifests

Cross-platform prerequisites (all platforms)

- Hardware/software/database requirements from Section 3 met and validated.
- DNS name resolved and (if applicable) reverse-proxy/Ingress configured (Section 6).
- Valid CAST Imaging license file available (Section 11).
- Outbound access to extend.castsoftware.com confirmed (Section 8–9).
- Authentication method decided (local / SAML / LDAP) and prerequisites gathered (Section 7).
- Firewall rules from the Network Ports Matrix (Section 10) opened before install day.

6. HTTPS / SSL

Reference: [install/https-ssl/](#)

Customer responsibility. CAST Imaging does not issue public TLS certificates. The customer must provide a valid certificate (from an internal or public Certificate Authority) and terminate/configure TLS, typically via a reverse proxy placed in front of CAST Imaging.

Prerequisites

- A certificate + private key (PEM/PFX) issued for the fully-qualified domain name (FQDN) under which CAST Imaging will be published.
- A reverse proxy or load balancer capable of TLS termination — e.g. Nginx, Apache HTTPD, HAProxy, F5, IIS with ARR, or a Kubernetes Ingress controller — sitting in front of the CAST Imaging web front end.
- Decision on TLS termination point: at the reverse proxy (recommended, offloads TLS from the application containers) vs. pass-through to the CAST Imaging web container itself.
- Certificate renewal process defined (manual or automated, e.g. via an internal PKI or ACME/Let's Encrypt where externally facing).
- HTTP → HTTPS redirect configured; disable plain HTTP once HTTPS is validated.

Typical topologies

1. **Reverse proxy in front of a single-machine Docker/Podman deployment** — proxy terminates TLS, forwards to the internal web service port over HTTP on a private network segment.
2. **Kubernetes Ingress** — Ingress controller (e.g. NGINX Ingress, Traefik) holds the TLS secret and routes to the CAST Imaging Service.
3. **Native Windows install** — IIS or a dedicated reverse proxy in front of the CAST Imaging web service.

7. Authentication

Reference: [install/authentication/](#), [authentication/local/](#), [authentication/saml/](#), [authentication/ldap/](#)

Method	Use case	Prerequisites
Local	POC/evaluation, small teams without centralized identity	None beyond the application itself; plan initial admin password rotation and account lifecycle management manually.
SAML 2.0	Enterprise SSO via a corporate Identity Provider (Azure AD/Entra ID, Okta, ADFS, PingFederate, etc.)	IdP metadata (or manual Entity ID / SSO URL / signing certificate); a registered Service Provider entry for CAST Imaging at the IdP; attribute/claim mapping (email, name, groups) agreed with the identity team; outbound/inbound HTTPS between browser, IdP, and CAST Imaging.
LDAP / Active Directory	Centralized authentication against an existing directory without full SSO	LDAP(S) server hostname/port, bind (service) account with read access, base DN(s) for user/group search, group-to-role mapping, LDAPS certificate trust if using TLS (port 636).

Cross-cutting prerequisites

- Decide role/permission mapping (admin, analyst, viewer) before go-live — especially for SAML/LDAP group-based mapping.
- Confirm network reachability: CAST Imaging → LDAP server (389/636) or CAST Imaging ⇄ IdP (443) as listed in the Network Ports Matrix (Section 10).
- Plan a break-glass local administrator account even when SAML/LDAP is the primary method.

8. CAST Extend Access

CAST Imaging relies on **CAST Extend** (extend.castsoftware.com) to download extensions/technology packages, product updates, and (depending on licensing model) license-related artifacts.

- Outbound HTTPS (443) from the CAST Imaging server to extend.castsoftware.com must be allowed (see Section 9 for the exact FQDN entry).
- An **API Key must be generated from the CAST Extend website** and configured in CAST Imaging (or the CAST Extend Local Update Server, if used) to authenticate extension downloads.
- For air-gapped/regulated environments without direct internet access, use a **CAST Extend Local Update Server (offline mirror)** deployed inside the network, which itself needs outbound access to extend.castsoftware.com and is the only component required to reach the internet.
- Proxy support: if outbound traffic must transit an authenticated HTTP(S) proxy, configure the proxy settings in CAST Imaging / CAST Extend client accordingly before first extension download.

Optional CAST Extend Local Update Server. Consider this component whenever individual CAST Imaging nodes must not have direct internet egress — it centralizes and audits all extension/update downloads through a single controlled outbound path.

9. Outbound Flows & Exact FQDN Whitelisting

No wildcard entries — allowlist the exact fully-qualified domain names below.

FQDN	Port	Purpose	Condition
<code>extend.castsoftware.com</code>	443	CAST Imaging extensions catalog, product updates, licensing artifacts	Mandatory
<code>cloud.casthighlight.com</code>	443	CAST Highlight SaaS Portal / MRI API integration	Conditional — Highlight integration
<code>auth.casthighlight.com</code>	443	CAST Highlight OAuth2 authentication endpoint	Conditional — Highlight integration
<code>casthighlight-binaries.s3.amazonaws.com</code>	443	Download of CAST Highlight analysis engines/binaries	Conditional — Highlight integration
<code>api.openai.com</code> (example)	443	LLM API endpoint for AI-assisted / MCP natural-language features	Conditional — AI/MCP, provider-specific
<code>api.anthropic.com</code> (example)	443	LLM API endpoint for AI-assisted / MCP natural-language features	Conditional — AI/MCP, provider-specific
Internal source repository hosts (Git/SVN)	22	SSH-based source code retrieval for analysis delivery	Conditional — Git/SVN over SSH
Internal file share hosts	445	SMB access to source code / delivery packages	Conditional — SMB-based delivery

No wildcards. Security policy requires each destination to be allowlisted by its exact FQDN (not a wildcard domain such as `*.amazonaws.com` or `*.openai.com`), since cloud-hosted endpoints share infrastructure with unrelated third-party services.

Full machine-readable version (includes inbound and internal flows too): [network-and-ports-matrix.csv](#).

10. Network Ports Matrix

Consolidated inbound, internal, and outbound port requirements. Full CSV export: [network-and-ports-matrix.csv](#)

Source	Destination	Port	Protocol	Reason	
End-user browser	CAST Imaging Web	443	HTTPS	Console / Viewer UI access	Mandatory
End-user browser	CAST Imaging Web	80	HTTP	Redirect to HTTPS only	Disable in prod
AIP Console / delivery tool	CAST Imaging	443/80	HTTPS/HTTP	Application package delivery	Mandatory
Admin workstation	CAST Imaging host OS	22 / 3389	SSH / RDP	Server administration	Mandatory
kubectl / CI	Kubernetes API server	6443	TCP	Cluster administration	K8s only
CAST Imaging app	PostgreSQL DB	5432	TCP	Application ⇌ database	Mandatory
CAST Imaging app	SQL Server DB	1433	TCP	Alternate database engine	If SQL Serv
Container host	Image registry	443	HTTPS	Pull CAST Imaging images	Mandatory
CAST Imaging / MCP	extend.castsoftware.com	443	HTTPS	Extensions, updates, license artifacts	Mandatory
CAST Imaging	cloud.casthighlight.com	443	HTTPS	Highlight SaaS integration	Conditional

Source	Destination	Port	Protocol	Reason	
CAST Imaging	auth.casthighlight.com	443	HTTPS	Highlight OAuth2	Conditional
CAST Imaging	casthighlight-binaries.s3.amazonaws.com	443	HTTPS	Highlight engine binaries	Conditional
MCP Server	LLM API endpoint	443	HTTPS	AI-assisted MCP responses	Conditional
MCP Gatekeeper	Identity Provider (OAuth/OIDC)	443	HTTPS	MCP client token validation	Conditional
CAST Imaging	Source repository (Git/SVN)	22	SSH	Source retrieval	Conditional
CAST Imaging	Source repository (HTTPS)	443	HTTPS	Source retrieval	Conditional
CAST Imaging	File share (SMB)	445	SMB	Source/package access	Conditional
CAST Imaging	SAML Identity Provider	443	HTTPS	SSO metadata / assertions	Conditional
CAST Imaging	LDAP / AD server	389 / 636	TCP / LDAPS	Directory authentication	Conditional
CAST Imaging	SMTP relay	25 / 587 / 465	TCP	Notification emails	Optional
CAST Imaging	NTP server	123	UDP	Time sync	Recommend
CAST Imaging	DNS server	53	UDP/TCP	Name resolution	Mandatory

11. Licensing

- A valid **CAST Imaging license file** must be obtained from CAST Software (via your CAST account team or CAST Extend) and supplied during installation/configuration.
- The license governs entitlement scope — e.g. number of applications, users, or modules (including whether MCP/AI features and Highlight integration are enabled).
- Plan the license renewal date into operational calendars; expired licenses can block analysis delivery or console access depending on grace-period configuration.
- An **API key generated from the CAST Extend website** is required separately from the license file, to authenticate automated extension/update downloads (see Section 8).
- Keep both the license file and API key in a secure secrets store (e.g. Kubernetes Secret, vault, encrypted file) rather than embedded in plain configuration files under version control.

Two distinct credentials. Do not confuse the **license file** (product entitlement) with the **CAST Extend API key** (download authentication) — both are required, obtained separately, and must be renewed/rotated independently.

12. Pre-Installation Checklist

Consolidated, printable checklist derived from Sections 2–11.

Architecture

- ☐ Platform selected (Windows / Docker / Podman / Kubernetes)

- ☐ Scenario selected (POC / Standard Production / Enterprise-HA)

- ☐ Topology selected (single-machine vs. multi-machine) and documented

Infrastructure

- ☐ Server(s) provisioned meeting CPU/RAM/disk sizing for the chosen scenario

- ☐ Database engine selected, sized, and provisioned (dedicated instance recommended)

- ☐ Persistent storage locations identified and separated (DB / logs / workspace / backup)

- ☐ DNS name(s) reserved and resolvable

- ☐ NTP time synchronization configured

Security & Network

- ☐ TLS certificate obtained; reverse proxy / Ingress selected and configured

- ☐ Authentication method decided (Local / SAML / LDAP) and IdP/LDAP prerequisites gathered

- ☐ All inbound/outbound firewall rules from the Network Ports Matrix opened

- ☐ Exact FQDNs allowlisted (no wildcards) for extend.castsoftware.com and any optional integrations

- ☐ Outbound proxy configuration (if applicable) validated for CAST Extend access

MCP Servers (if AI features enabled)

- ☐ MCP Server and Gatekeeper deployment planned alongside core platform

- ☐ OAuth2/OIDC provider identified for MCP client authentication

- ☐ LLM provider selected and its exact API FQDN allowlisted
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Licensing & Access

- ☐ CAST Imaging license file obtained and ready to install
 - ☐ CAST Extend API key generated and stored securely
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End Users

- ☐ End-user workstation requirements communicated (browser, RAM, resolution)
 - ☐ Delivery/analyst workstation requirements communicated (JRE, SCM clients, workspace disk)
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13. Appendix & References

Machine-readable exports

- [network-and-ports-matrix.csv](#) — full inbound / internal / outbound port and FQDN matrix
- [sizing-requirements-matrix.csv](#) — hardware sizing matrix by scenario and topology

Source documentation structure referenced

- install/before-you-start/deployment-options/
- install/requirements/ (hardware, software, db, disk, disk/storage-locations)
- imaging/mcp-server/ (imaging, gatekeeper, oauth)
- install/global/ (windows, docker, podman, kubernetes)
- install/https-ssl/
- install/authentication/ (local, saml, ldap)

Reminder. Verify every version number, minimum requirement, and default port in this document against the live doc.castsoftware.com/imaging/install portal for the CAST Imaging release you are about to deploy — this document is a planning aid, not a substitute for the official release notes and installation guide.